



# Pistil: A 16-nm Accelerator Co-Designed with a 20-Chiplet 2.5D System-in-Package Architecture for Distributed Small Language Model Inference at the Edge

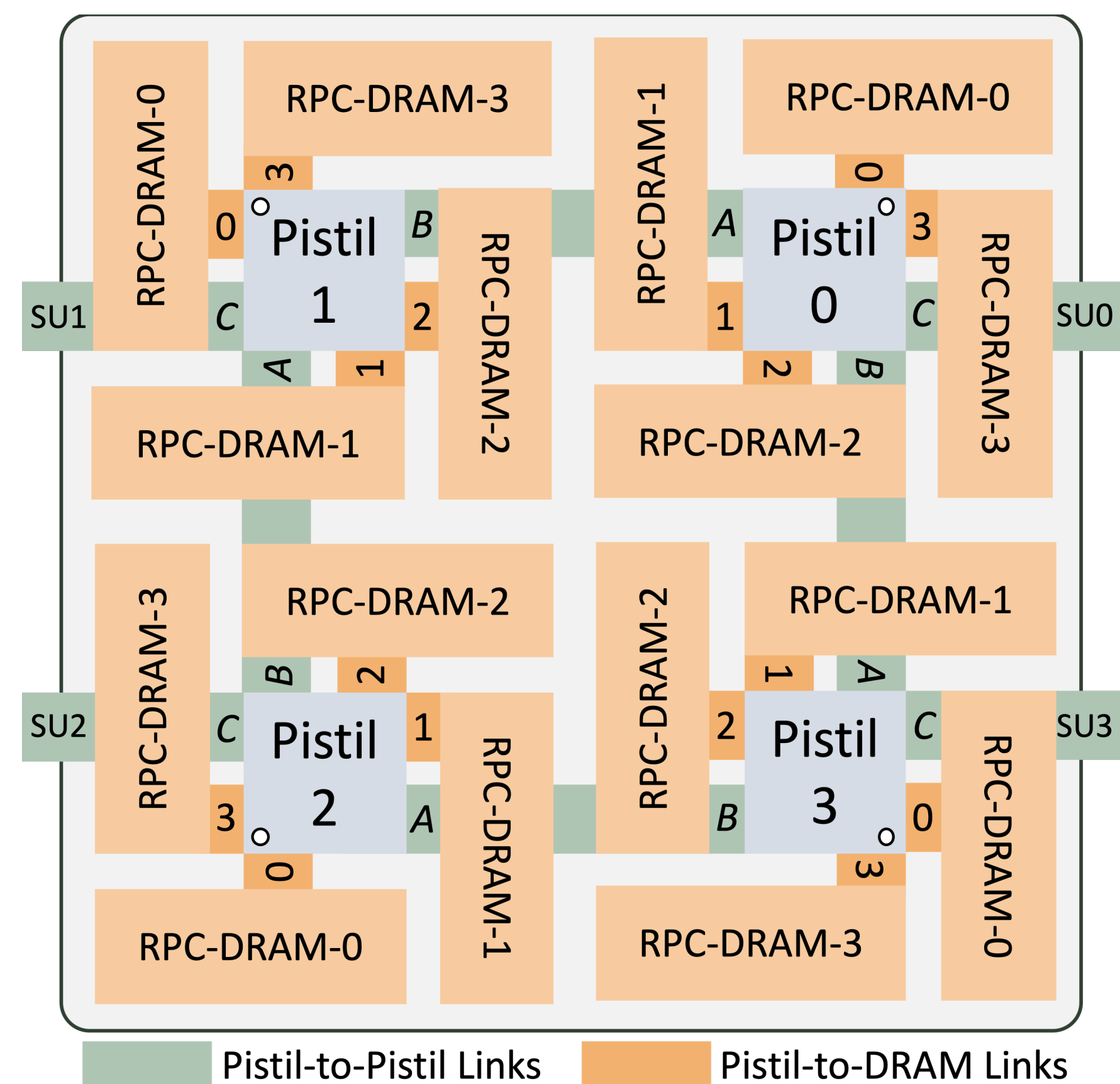


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## Motivation

### SLM decode is a memory-bandwidth problem.

Dense small language model decode is dominated by repeated vector-matrix multiplication over model weights. Because decode generates tokens sequentially, each new token requires repeatedly streaming weights from memory, making latency and energy depend heavily on sustained memory bandwidth rather than peak compute alone. For edge systems, memory capacity, bandwidth, cost, and energy must be balanced carefully. Extra capacity does not improve decode performance unless the architecture can also deliver enough bandwidth to keep compute active. Pistil targets this bottleneck by co-designing RPC-DRAM memory chiplets, compute chiplets, package topology, and streaming execution to maximize bandwidth per capacity for low-latency, energy-efficient SLM inference.



Multi-chiplet system-in-package with four Pistil compute chiplets and sixteen RPC-DRAM memory chiplets.

### Key Idea: Fully utilize memory bandwidth – the key resource – while compute and network operate independently.

Pistil uses a chiplet-independent execution model with **no shared memory address space**. Each Pistil chiplet decouples memory, compute, and network execution into independent pipelines, each with its own instruction queue and decode frontend. Hardware-managed synchronization through scratchpads allows memory streaming, computation, and chiplet-to-chiplet communication to overlap at fine granularity.

## Architecture and Prototype

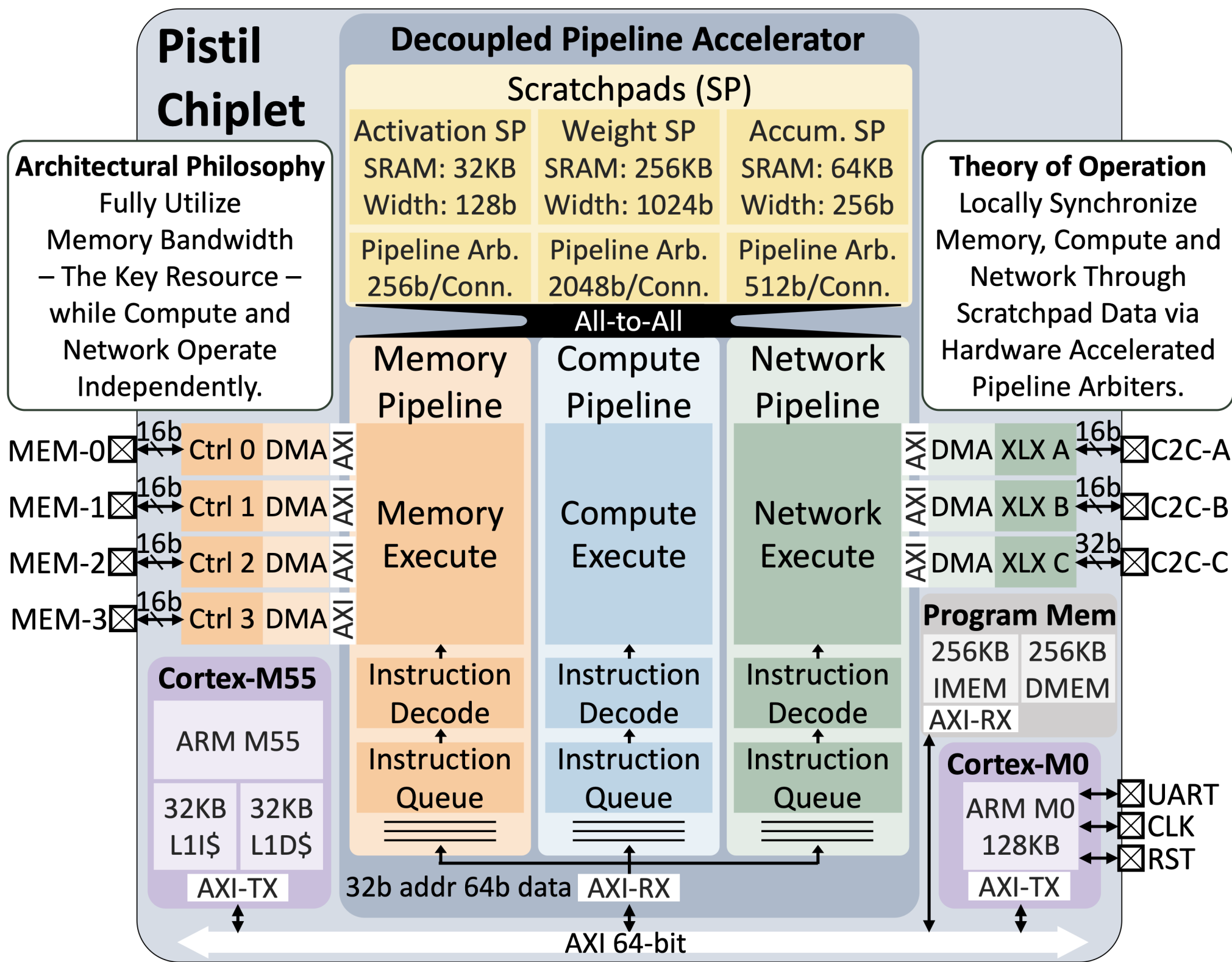
### Decoupled execution model

Pistil separates accelerator execution into three independent pipelines:

**Memory pipeline:** Streams weights and activations from RPC-DRAM into scratchpads.

**Compute pipeline:** Consumes scratchpad-resident tensors for VMM, vector operations, and accumulation.

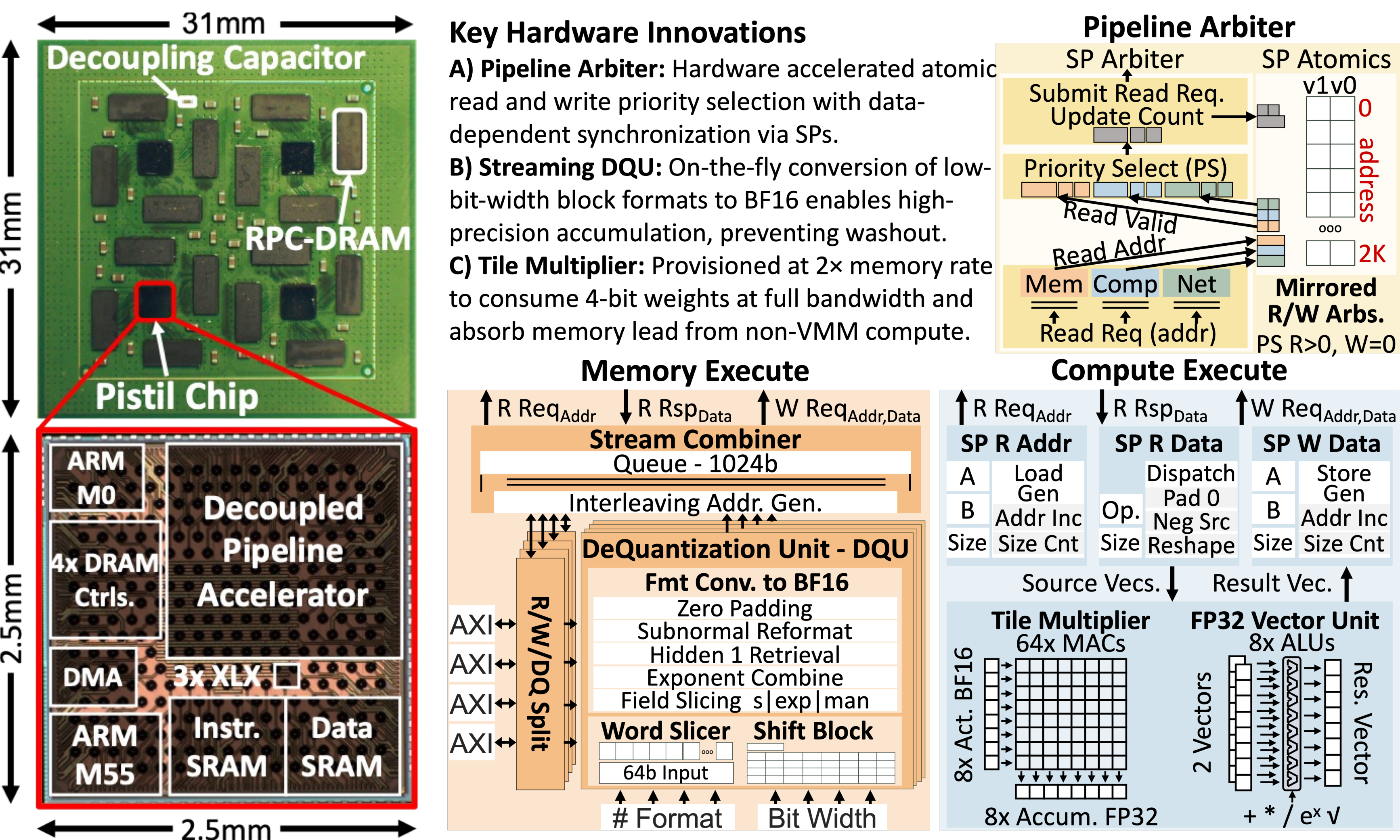
**Network pipeline:** Moves data across Pistil chiplets through custom chip-to-chip links.



A lightweight Arm Cortex-M55 issues coarse-grained commands that configure internal hardware FSMs, enabling concurrent memory movement, computation, and communication without explicit micro-kernel interleaving.

### System architecture

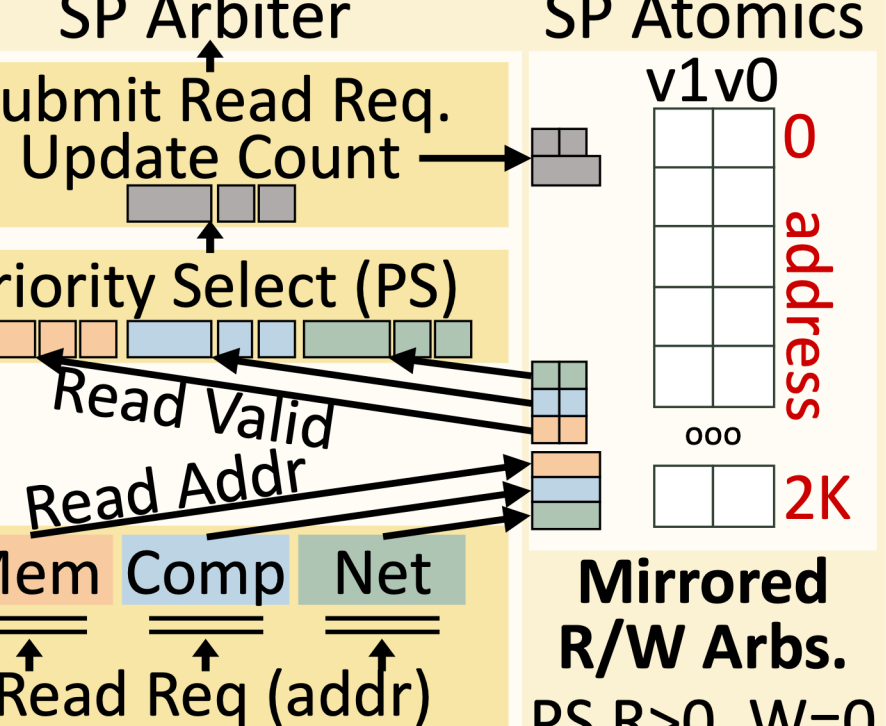
Pistil is a 2.5D system-in-package for distributed edge SLM inference. The system integrates 4 Pistil compute chiplets with 16 RPC-DRAM memory chiplets, providing 512 MB of total DRAM capacity and 51.2 GB/s of modeled total DRAM bandwidth. The package supports model sizes up to 1B parameters and uses custom Pistil-to-Pistil package links, along with scale-up links for board-level communication. The package is organized using a modular “flower” topology, where each Pistil compute chiplet is surrounded by four RPC-DRAM chiplets. This topology efficiently uses compute-memory shoreline and allows memory capacity and bandwidth to scale with the number of flowers in the system.



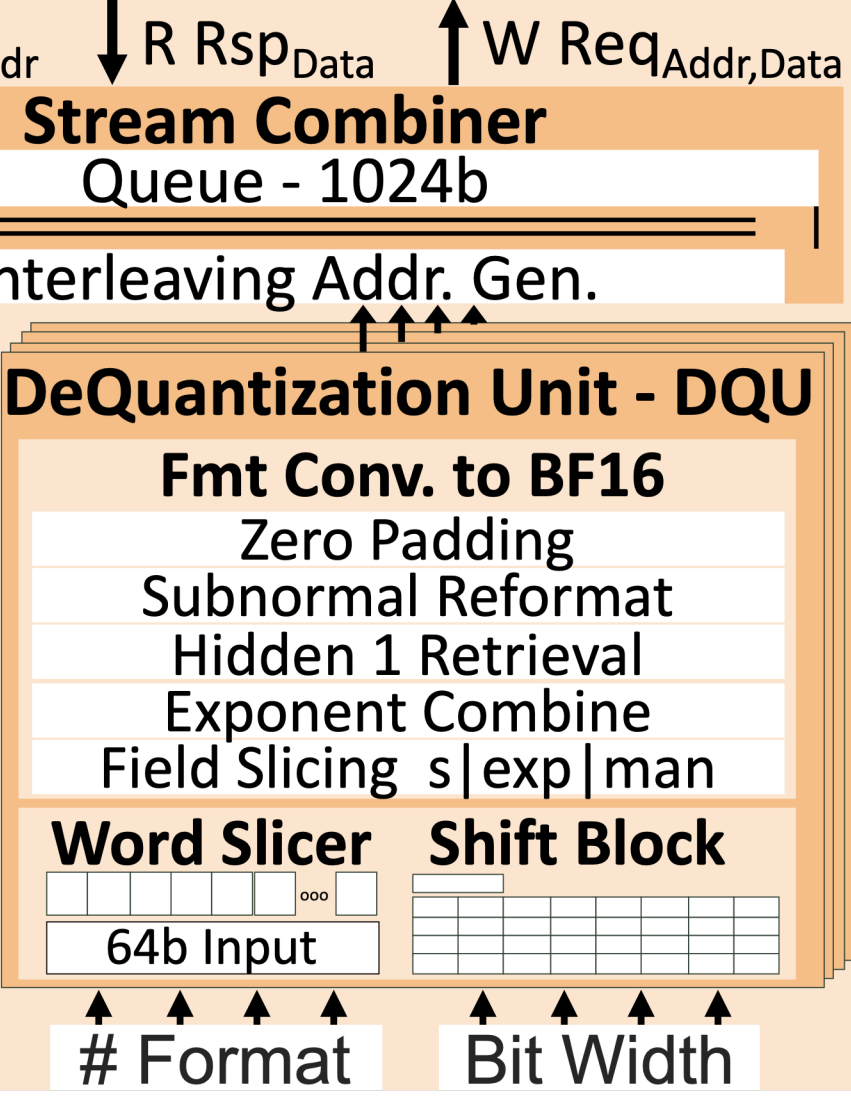
### Key Hardware Innovations

- A) Pipeline Arbiter:** Hardware accelerated atomic read and write priority selection with data-dependent synchronization via SPs.
- B) Streaming DQU:** On-the-fly conversion of low-bit-width block formats to BF16 enables high-precision accumulation, preventing washout.
- C) Tile Multiplier:** Provisioned at 2x memory rate to consume 4-bit weights at full bandwidth and absorb memory lead from non-VMM compute.

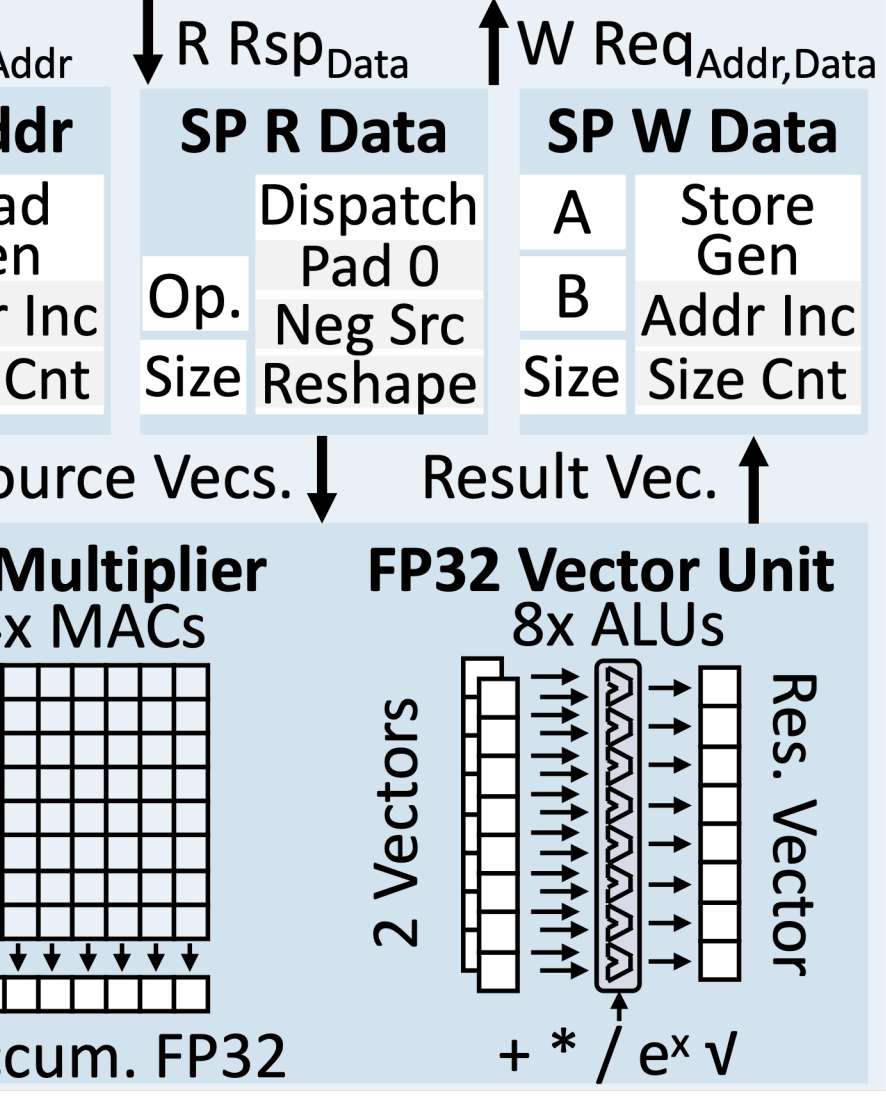
### Pipeline Arbiter



### Memory Execute

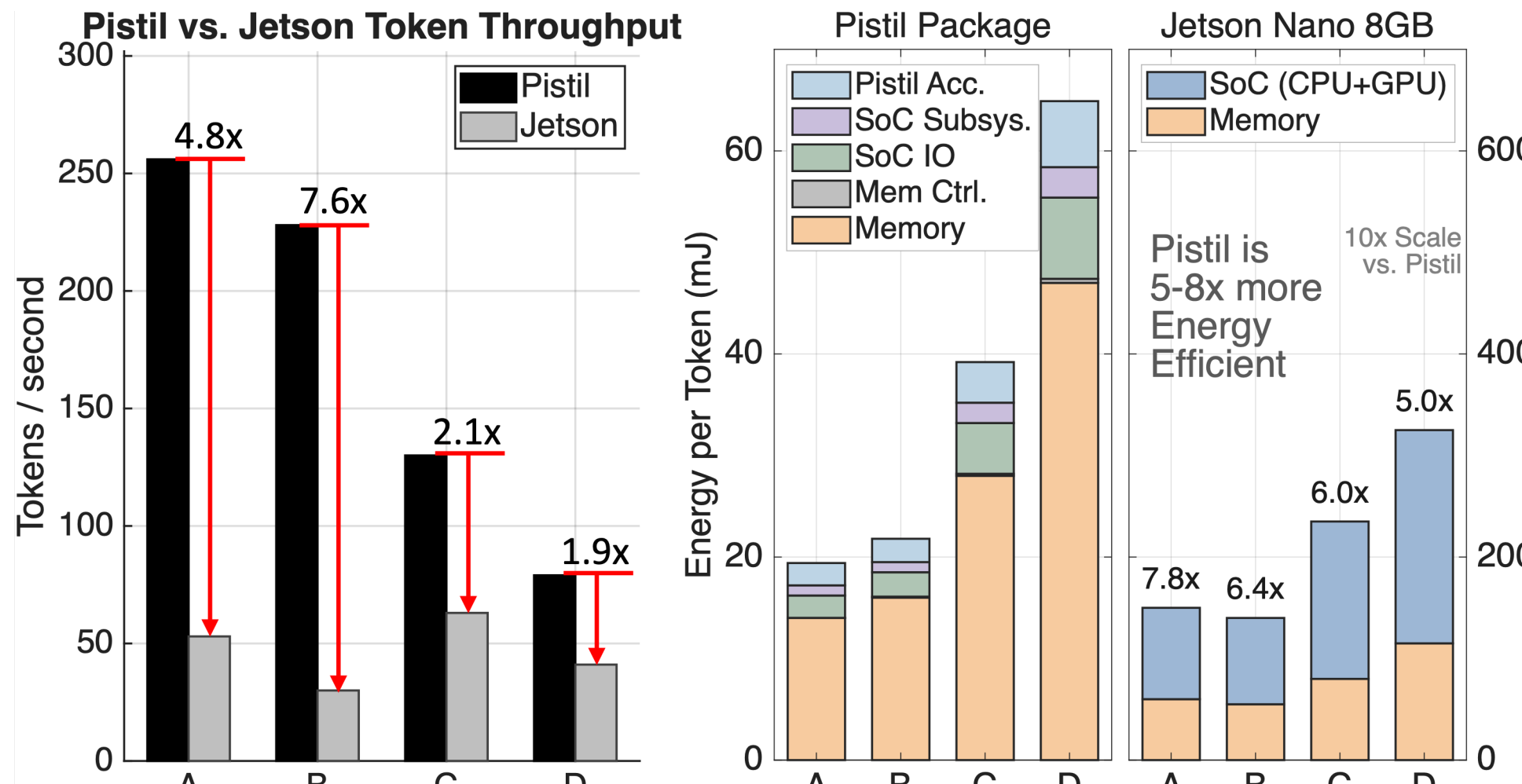


### Compute Execute



## Measured Results

Across representative dense SLM workloads, Pistil achieves **80–250 tokens/s, up to 7.6× higher token throughput than Jetson Nano 8GB**, and **over 98% of peak memory bandwidth during decode**. Pistil also achieves **5.0-7.8× lower energy per token than Jetson Nano 8GB**, demonstrating that sustained memory bandwidth is the dominant driver of efficient edge SLM decode. These gains come from Pistil’s ability to continuously stream weights, perform on-the-fly dequantization, and overlap memory access, computation, and inter-chiplet communication at fine granularity.



| Model        | Model A<br>Gemma-3 270M | Model B<br>Granite-4 350M | Model C<br>Qwen-3 600M | Model D<br>Gemma-3 1B |
|--------------|-------------------------|---------------------------|------------------------|-----------------------|
| Num. Layers  | 18                      | 28                        | 28                     | 26                    |
| Hidden Dim   | 640                     | 1024                      | 1024                   | 1152                  |
| MLP Inter.   | 2048                    | 2048                      | 3072                   | 6912                  |
| Atten. Heads | 4Q / 1KV                | 16Q / 4KV                 | 16Q / 8KV              | 4Q / 1KV              |
| MLP Act.     | GeLU                    | SiLU                      | SiLU                   | GeLU                  |
| Vocab Size   | 262144                  | 100352                    | 151936                 | 262144                |
| Capacity     | 154 MB                  | 207 MB                    | 330 MB                 | 511 MB                |

Pistil SiP vs. Jetson Nano 8GB decoding performance with llama.cpp 4-bit quantization.

|                                | HCS'22 [1]<br>Jetson | ISSCC'25 [2]<br>Slim-Llama       | VLSI'25 [3]<br>Adelia | This Work<br>Pistil        |
|--------------------------------|----------------------|----------------------------------|-----------------------|----------------------------|
| Technology                     | 20nm                 | 28nm                             | 4nm                   | 16nm                       |
| LLM Phase Opt.                 | Prefill              | Prefill                          | Prefill               | Decode                     |
| Supply Volt. [V]               | 0.5 – 1.1            | 0.58 – 1.0                       | 0.65 – 0.9            | 0.8                        |
| Frequency [MHz]                | 640 – 1020           | 25 – 200                         | 25 – 1000             | 25 – 800 <sup>d</sup>      |
| # Logic Chiplets               | 1                    | 1                                | 1                     | 4                          |
| SiP Logic Area                 | 118mm <sup>2</sup>   | 20.25mm <sup>2</sup>             | 5.28mm <sup>2</sup>   | 25mm <sup>2</sup>          |
| Power [mW]                     | 10W – 25W            | 4.69 – 82.07                     | 11 – 735              | 20-1000 <sup>d</sup>       |
| Data Types                     | Int8/BF16            | A: Int 4/8/16<br>W: Int1-16/Tern | FP16                  | A: BF16/FP32<br>W: MXFP4-8 |
| Compute (TOPs)                 | 33.5                 | 4.9 – 13.1                       | 8.19                  | 0.4                        |
| Memory to Logic Shoreline [mm] | 10mm                 | 2mm                              | 10mm                  | 40mm                       |
| Mem Tech.                      | LP-DDR5              | DDR3                             | Simulated<br>HBM2E    | RPC-DRAM                   |
| Capacity [GB]                  | 8                    | 3 <sup>a</sup>                   |                       | 0.5                        |
| Mem BW [GB/s]                  | 102                  | 1.6                              |                       | 51.2 <sup>d</sup>          |
| Gemma3 Tok/s                   | 40.6                 | 3.2                              |                       | 79 <sup>d</sup>            |
| 1B <sup>c</sup> mJ/Tok         | 325                  | 77.4 <sup>b</sup>                |                       | 64.5 <sup>d</sup>          |

<sup>a</sup>From Max Model Capacity. <sup>b</sup>8-bit weights w/ EMA. <sup>c</sup>Decode Phase 256 sequence length. <sup>d</sup>Measured up to 50MHz and modeled at 800MHz.

### Main takeaway

Pistil shows that edge SLM decode performance is best improved by sustaining memory bandwidth. By combining RPC-DRAM chiplets, a 2.5D package topology, decoupled pipelines, scratchpad-based synchronization, and streaming dequantization, Pistil delivers high-throughput, energy-efficient dense SLM inference in an edge-class system.